

## Asymptotic models for thin transversely isotropic elastic layers\*

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**Abstract** Thin elastic layers bonded on one or both surfaces to rigid substrates are ubiquitous in coatings, adhesives, and integrated circuits. Their response to applied surface tractions is commonly represented by reduced mattress (or Winkler) models and shear-lag models. Classical reduced relations typically ignore normal-tangential coupling and are often confined to compressible and isotropic solids. In this work, we develop a Hankel-transform formulation for axisymmetric loading of transversely isotropic layers and perform a systematic thin-layer (or long-wavelength) asymptotic expansion to obtain high-order surface traction-displacement relations. The resulting reduced relations provide extended Winkler and shear-lag-type models that retain the coupling between normal and shear responses and accommodate transverse isotropy. These asymptotic models provide a practical basis for predicting deformation fields and stress transfer in layered and bonded systems.

**Key words** transversely isotropic elasticity, thin layer, Winkler foundation, shear-lag, contact mechanics, asymptotic method

**Chinese Library Classification** O343.2

**2020 Mathematics Subject Classification** 74K35

## 1 Introduction

Thin elastic layers with one or both of their surfaces confined are ubiquitous in nature and engineering systems, ranging from thin coatings to adhesive joints<sup>[1–3]</sup>. An important mechanical feature of such layers is that their three-dimensional elasticity can often be reduced, through appropriate approximations over the thickness direction, to effective two-dimensional models<sup>[4]</sup>. Specifically, the layer response to external loading is frequently represented as an array of independent linear springs, giving rise to the classical Winkler foundation model for normal loading<sup>[5]</sup> and the shear-lag model for tangential loading<sup>[6]</sup>. Both models have proven particularly useful and versatile for describing stress transfer and deformation in bonded or layered systems<sup>[4]</sup>. A concrete example could be given by their uses for the seminal analysis

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on single-lap joints<sup>[7]</sup>, which not only established a standard configuration for testing adhesive joints, but also inspired a broad class of analytical models for elastohydrodynamic lubrication, adhesion, and composite mechanics<sup>[8–10]</sup>.

Despite their success and enduring appeal, these classical models rely on simplifying assumptions, such as material compressibility and the decoupling of normal and shear responses that limit their range of applicability<sup>[4]</sup>. To overcome these limitations, heuristic approaches such as Johnson’s “plane sections remain plane after compression” assumption<sup>[1]</sup> are not applicable. In contrast, a formal asymptotic analysis that systematically exploits the slenderness of the layer provides a rigorous route to extend classical foundation and shear-lag models, enabling the inclusion of higher-order (incompressibility) and coupling effects<sup>[11]</sup>. Examples include higher-order foundation models for thin layers under sliding contact with a substrate (also known as the Kerr foundation model<sup>[11]</sup>), perfectly bonded layers (the modified Kerr foundation model<sup>[12–13]</sup>), perfectly incompressible thin layers<sup>[14–15]</sup>, arbitrarily incompressible/compressible layers<sup>[16–17]</sup>, and coupled foundation-shear-lag formulations<sup>[18]</sup>. It has become clear that higher-order modeling is essential for accurately describing elastomeric systems, where Poisson’s ratio is very close to  $1/2$ <sup>[16–19]</sup>.

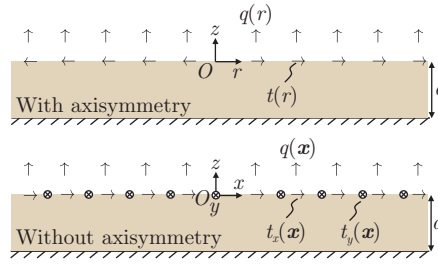
In a wide range of material systems, including laminated composites, dielectric films, anisotropic coatings, biological tissues, and liquid-crystal elastomers<sup>[20–21]</sup>, the elastic layer is more accurately described as transversely isotropic<sup>[22–24]</sup>. This inherent anisotropy fundamentally modifies both the normal and tangential compliances, as well as their coupling behavior. Although high-order or coupled foundation models have been carefully developed for isotropic elastic layers, analogous formulations for transversely isotropic layers remain scarce. Existing analyses often rely on ad hoc simplifications<sup>[25]</sup> or are restricted to specific loading configurations, particularly frictionless contact without any shear surface tractions<sup>[26–29]</sup>. A unified theoretical framework, which rigorously links the three-dimensional elasticity of transversely isotropic layers to effective two-dimensional surface relations and consistently incorporates coupling between normal and tangential fields, remains elusive, as do coupled higher-order foundation and shear-lag models.

The objective of this work is to develop a unified theoretical framework for describing the deformation of transversely isotropic elastic layers under the thin-layer (or long-wavelength) assumption. Section 2 presents the Hankel-transform formulation and derives the corresponding surface compliance and stiffness relations. Sections 3 and 4 discuss the asymptotic results in the long-wavelength limit for perfectly bonded interfaces and frictionless interfaces, respectively. Section 5 consolidates these results into generalized Winkler- and shear-lag-type models, and examines their isotropic limits. Finally, Section 6 summarizes the key findings and discusses potential extensions to the contact, adhesion, and indentation problems.

## 2 A transform approach

Figure 1 shows both an axisymmetric  $(r, z)$  meridional sketch (used for the derivation) and a schematic written in terms of the in-plane position  $\mathbf{x} = (x, y)$ . For convenience, we formulate the transversely isotropic elastic-layer problem in the cylindrical coordinates  $(r, z)$  under axisymmetry (see the upper panel of Fig. 1), which yields a compact Hankel-domain representation, where all field quantities depend only on the radial coordinate  $r$  and the thickness coordinate  $z$ . The extension to the general three-dimensional case is obtained later by matching the corresponding in-plane operators. Accordingly, the applied normal traction is written as  $q(r)$  under axisymmetry but as  $q(\mathbf{x})$  in the general in-plane setting.

Under axisymmetry, there is no circumferential variation, and the azimuthal displacement vanishes. The nonzero displacement components are therefore the radial and axial displacements, i.e.,



**Fig. 1** Schematic cross-section of a transversely isotropic elastic layer of thickness  $d$  on a rigid substrate. The layer is subject to a normal load and an in-plane shear traction on its upper surface ( $z = 0$ ). The top and bottom panels show notations with axisymmetry and without axisymmetry, respectively (color online)

$$u_r = u_r(r, z), \quad u_z = u_z(r, z). \quad (1)$$

The strain components associated with these displacements are

$$\varepsilon_{rr} = \frac{\partial u_r}{\partial r}, \quad \varepsilon_{\theta\theta} = \frac{u_r}{r}, \quad \varepsilon_{zz} = \frac{\partial u_z}{\partial z}, \quad \varepsilon_{rz} = \frac{1}{2} \left( \frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r} \right). \quad (2)$$

Correspondingly, the nonzero stress components in the cylindrical coordinates  $(r, z)$  are  $\sigma_{rr}$ ,  $\sigma_{\theta\theta}$ ,  $\sigma_{zz}$ , and  $\sigma_{rz} = \sigma_{zr}$ , and the equilibrium equations in the absence of body forces reduce to

$$\frac{\partial \sigma_{rr}}{\partial r} + \frac{\partial \sigma_{rz}}{\partial z} + \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} = 0, \quad \frac{\partial \sigma_{rz}}{\partial r} + \frac{\partial \sigma_{zz}}{\partial z} + \frac{\sigma_{rz}}{r} = 0. \quad (3)$$

To represent these relationships in a compact analytical form<sup>[30]</sup>, a Love stress function  $\Phi(r, z)$  can be introduced such that the stress components can be written as<sup>[31]</sup>

$$\begin{cases} \sigma_{rz} = \frac{\partial}{\partial r} \left( \frac{\partial^2}{\partial r^2} + \frac{\partial}{r \partial r} - \frac{C_{13}}{C_{11}} \frac{\partial^2}{\partial z^2} \right) \Phi, \\ \sigma_{zz} = -\frac{\partial}{\partial z} \left( \frac{C_{11}C_{33} - C_{13}^2 - C_{44}C_{13}}{C_{11}C_{44}} \left( \frac{\partial^2}{\partial r^2} + \frac{\partial}{r \partial r} \right) + \frac{C_{33}}{C_{11}} \frac{\partial^2}{\partial z^2} \right) \Phi, \\ \sigma_{rr} = -\frac{\partial}{\partial z} \left( \frac{\partial^2}{\partial r^2} + \frac{C_{12}C_{44} - C_{11}C_{13} + C_{12}C_{13}}{C_{11}C_{44}} \frac{\partial}{r \partial r} - \frac{C_{13}}{C_{11}} \frac{\partial^2}{\partial z^2} \right) \Phi, \\ \sigma_{\theta\theta} = -\frac{\partial}{\partial z} \left( \frac{C_{12}C_{44} - C_{11}C_{13} + C_{12}C_{13}}{C_{11}C_{44}} \frac{\partial^2}{\partial r^2} + \frac{\partial}{r \partial r} - \frac{C_{13}}{C_{11}} \frac{\partial^2}{\partial z^2} \right) \Phi. \end{cases} \quad (4)$$

The material has five independent elastic constants  $(C_{11}, C_{12}, C_{13}, C_{33}, C_{44})$ , or equivalently  $(E, E', \nu, \nu', G')$ , related by

$$\begin{cases} C_{11} = \frac{EE' - E^2\nu'^2}{E'(1 - \nu^2) - 2E\nu'^2(1 + \nu)}, & C_{12} = \frac{EE'\nu + E^2\nu'^2}{E'(1 - \nu^2) - 2E\nu'^2(1 + \nu)}, \\ C_{13} = \frac{EE'\nu'}{E'(1 - \nu) - 2E\nu'^2}, & C_{33} = \frac{E'^2(1 - \nu)}{E'(1 - \nu) - 2E\nu'^2}, \quad C_{44} = G'. \end{cases} \quad (5)$$

Here,  $E$  and  $E'$  are Young's moduli in the symmetry plane and along the symmetry axis, respectively;  $\nu$  and  $\nu'$  are the corresponding Poisson's ratios;  $G'$  is the shear modulus in the plane normal to the axis. Note that  $C_{66}$  is often used in the numerical calculation for convenience, which reads  $C_{66} = E/(2(1 + \nu))$ . These constants reduce to the isotropic form when  $E' = E$ ,  $G' = E/(2(1 + \nu))$ , and  $\nu' = \nu$ .

Thus, the radial equilibrium condition (i.e., the first equation in Eq. (3)) is automatically satisfied. The remaining equilibrium equation in Eq. (3) then requires  $\Phi(r, z)$  to satisfy a generalized biharmonic equation of the form

$$\left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{s_1^2} \frac{\partial^2}{\partial z^2}\right) \left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{s_2^2} \frac{\partial^2}{\partial z^2}\right) \Phi = 0, \quad (6)$$

where the parameters  $s_1$  and  $s_2$  are given by

$$s_{1,2}^2 = \frac{C_{11}C_{33} - 2C_{13}C_{44} - C_{13}^2 \pm \sqrt{(C_{11}C_{33} - C_{13}^2)^2 - 4(C_{11}C_{33} - C_{13}^2)(C_{13}C_{44} + C_{44}^2)}}{2C_{33}C_{44}}. \quad (7)$$

This has been shown and proven by Ref. [30] in Eq. (78.16) (where the theorem is right below) that  $s_1$  and  $s_2$  for any transversely isotropic elastic body can only be real or complex (with a real part different from 0). For example, using representative engineering constants for unidirectional carbon/epoxy (IM7/977-3) and graphite/epoxy (GY-70/934) laminas (see Table A.4, Page 378 in Ref. [32]), we obtain  $s_1^2 = 1.3409$ ,  $s_2^2 = 0.0439$  for IM7/977-3 and  $s_1^2 = 1.3360$ ,  $s_2^2 = 0.0171$  for GY-70/934.

In the isotropic limit  $s_1 = s_2 = 1$ , Eq. (6) reduces to the classical biharmonic equation governing axisymmetric deformations of isotropic elastic solids<sup>[31]</sup>. In addition, the strain-displacement compatibility conditions are automatically satisfied by defining the displacement and stress components in terms of the stress function  $\Phi(r, z)$  as

$$u_z = \left(\frac{1}{C_{44}} \frac{\partial^2}{\partial r^2} + \frac{1}{C_{44}} \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{C_{11}} \frac{\partial^2}{\partial z^2}\right) \Phi, \quad u_r = \frac{C_{13} + C_{44}}{C_{11}C_{44}} \frac{\partial^2 \Phi}{\partial r \partial z}. \quad (8)$$

Similar to the isotropic case<sup>[31]</sup>, the solution to the transversely isotropic problem described by Eq. (6) can be obtained by employing the Hankel transform technique for axisymmetric systems<sup>[33]</sup>. Specifically, the  $m$ th order Hankel transform of the radially dependent function  $f(r, z)$  is defined as

$$\tilde{f}(\xi, z) = \int_0^\infty r f(r, z) J_m(\xi r) dr, \quad (9)$$

and the corresponding inverse transform is

$$f(r, z) = \int_0^\infty \xi \tilde{f}(\xi, z) J_m(\xi r) d\xi, \quad (10)$$

where  $J_m(\cdot)$  denotes the Bessel function of the first kind of order  $m$ . A useful property of the Hankel transform, frequently employed in the subsequent analysis, is

$$\int_0^\infty r \left( \frac{d^2 f}{dr^2} + \frac{1}{r} \frac{df}{dr} - \frac{m^2}{r^2} f \right) J_m(\xi r) dr = -\xi^2 \tilde{f}(\xi), \quad (11)$$

and thus all partial differential relations with respect to  $(r, z)$  are reduced to ordinary differential relations in the independent variable  $z$ .

By applying the zeroth-order Hankel transform ( $m = 0$ ) to Eq. (6), the governing equation is converted into an ordinary differential equation in  $z$ ,

$$\left(\frac{d^2}{s_1^2 dz^2} - \xi^2\right) \left(\frac{d^2}{s_2^2 dz^2} - \xi^2\right) \tilde{\Phi}(\xi, z) = 0, \quad (12)$$

where the transformed potential  $\tilde{\Phi}(\xi, z)$  is defined by the zeroth-order Hankel transform, i.e.,

$$\tilde{\Phi}(\xi, z) = \int_0^\infty r \Phi(r, z) J_0(\xi r) dr. \quad (13)$$

Equation (12) is a fourth-order ordinary differential equation in  $z$  for each wavenumber  $\xi$ , and its general solution can be written as

$$\tilde{\Phi}(\xi, z) = A_1 \cosh(s_1 \xi z) + A_2 \cosh(s_2 \xi z) + A_3 \sinh(s_1 \xi z) + A_4 \sinh(s_2 \xi z). \quad (14)$$

Note that in the isotropic limit, where  $s_1 = s_2 = 1$ , the solution reduces to the classical form<sup>[31]</sup>,

$$\tilde{\Phi}(\xi, z) = (A_1 + A_2 z) \cosh(\xi z) + (A_3 + A_4 z) \sinh(\xi z). \quad (15)$$

Equation (14) has been previously simplified for transversely isotropic half-spaces<sup>[30]</sup>. There is also discussion for the case of a finite thickness with a lengthy expression for the constants<sup>[33–35]</sup>. In the present study, however, we are concerned with a finite elastic layer and seek simplification under the assumption of a small thickness. In this case, the constants  $A_i$  are determined by four boundary conditions, which depend on how the layer is constrained. These boundary conditions typically involve combinations of stress and displacement components, whose relations to the transformed potential  $\tilde{\Phi}$  can be obtained by applying the Hankel transform, followed by the inverse transform, to the equilibrium and compatibility relations given in Eqs. (4)–(8).  $\tilde{\sigma}_{zz}(\xi, z)$  and  $\tilde{u}_z(\xi, z)$  are defined by the zeroth-order Hankel transform, while  $\tilde{\sigma}_{rz}(\xi, z)$  and  $\tilde{u}_r(\xi, z)$  are defined by the first-order one. Explicitly, the relevant field quantities are expressed as

$$\begin{cases} \tilde{\sigma}_{zz}(\xi, z) = \frac{C_{11}C_{33} - C_{13}^2 - C_{13}C_{44}}{C_{11}C_{44}} \xi^2 \frac{d\tilde{\Phi}}{dz} - \frac{C_{33}}{C_{11}} \frac{d^3\tilde{\Phi}}{dz^3}, \\ \tilde{\sigma}_{rz}(\xi, z) = \xi^3 \tilde{\Phi} + \frac{C_{13}}{C_{11}} \xi \frac{d^2\tilde{\Phi}}{dz^2}, \\ \tilde{u}_z(\xi, z) = \frac{1}{C_{11}} \frac{d^2\tilde{\Phi}}{dz^2} - \frac{1}{C_{44}} \xi^2 \tilde{\Phi}, \\ \tilde{u}_r(\xi, z) = -\frac{C_{13} + C_{44}}{C_{11}C_{44}} \xi \frac{d\tilde{\Phi}}{dz} \end{cases} \quad (16)$$

in the Hankel space, and

$$\begin{cases} \sigma_{zz}(r, z) = \int_0^\infty \xi \left( \frac{C_{11}C_{33} - C_{13}^2 - C_{13}C_{44}}{C_{11}C_{44}} \xi^2 \frac{d\tilde{\Phi}}{dz} - \frac{C_{33}}{C_{11}} \frac{d^3\tilde{\Phi}}{dz^3} \right) J_0(\xi r) d\xi, \\ \sigma_{rz}(r, z) = \int_0^\infty \xi^2 \left( \xi^2 \tilde{\Phi} + \frac{C_{13}}{C_{11}} \frac{d^2\tilde{\Phi}}{dz^2} \right) J_1(\xi r) d\xi, \\ u_z(r, z) = \int_0^\infty \xi \left( \frac{1}{C_{11}} \frac{d^2\tilde{\Phi}}{dz^2} - \frac{1}{C_{44}} \xi^2 \tilde{\Phi} \right) J_0(\xi r) d\xi, \\ u_r(r, z) = -\frac{C_{13} + C_{44}}{C_{11}C_{44}} \int_0^\infty \xi^2 \frac{d\tilde{\Phi}}{dz} J_1(\xi r) d\xi \end{cases} \quad (17)$$

in the physical space. We next determine the coefficients  $A_i$  by enforcing specific boundary conditions at the lower surface of the layer, corresponding to either a frictionless (no-shear) or a perfectly bonded (no-slip) interface. These conditions lead to distinct forms of the traction-displacement or displacement-traction relations at the upper surface.

### 3 Perfectly bonded elastic layers

We first discuss the case of an elastic layer perfectly bonded at  $z = -d$  (see Fig. 1). In this case, both the vertical and radial displacements must vanish at the lower surface. At the upper surface, prescribed tractions are applied, and the resulting surface displacements are computed to establish the corresponding traction-displacement relations (or displacement-traction relations for prescribed displacements).

#### 3.1 Prescribed surface tractions

##### 3.1.1 Coupled and high-order solutions

We consider a prescribed normal load  $q(r)$  and a tangential load  $t(r)$  applied at the top surface of the layer. Let  $\tilde{q}(\xi)$  and  $\tilde{t}(\xi)$  denote, respectively, the zeroth-order Hankel transform of  $q(r)$  and the first-order Hankel transform of  $t(r)$ . The boundary conditions in the transformed domain are therefore written as

$$\begin{cases} \tilde{\sigma}_{zz}(\xi, 0) = \tilde{q}, & \tilde{\sigma}_{rz}(\xi, 0) = \tilde{t}, \\ \tilde{u}_z(\xi, -d) = 0, & \tilde{u}_r(\xi, -d) = 0, \end{cases} \quad (18)$$

which correspond to prescribed tractions at the upper surface and a perfectly bonded condition at the lower surface. Combining Eq. (18) with Eq. (16) allows the coefficients in Eq. (14) to be determined. Consequently, the surface displacements in the Hankel domain can be expressed in terms of the transformed tractions as

$$\tilde{u}_z(\xi, 0) = \mathcal{F}_{zz}(\xi) \tilde{q}(\xi) + \mathcal{F}_{zr}(\xi) \tilde{t}(\xi), \quad (19)$$

$$\tilde{u}_r(\xi, 0) = \mathcal{F}_{rz}(\xi) \tilde{q}(\xi) + \mathcal{F}_{rr}(\xi) \tilde{t}(\xi), \quad (20)$$

where  $\mathcal{F}_{ij}(\xi)$  ( $i, j = z, r$ ) are the surface compliance functions, and  $\mathcal{F}_{ij} = \mathcal{F}_{ji}$ . The symbolic code used to solve the corresponding linear system and generate  $\mathcal{F}_{ij}(\xi)$  is available online (see <https://github.com/14523GuozhengZhang/Asymptotic-models.git>).

Notably, for isotropic layers, Eq. (19) reduces to

$$\tilde{u}_z(\xi, 0) = \frac{(1 - \nu)((3 - 4\nu) \sinh(2d\xi) - 2d\xi)\tilde{q} - ((d\xi)^2 - (3 - 10\nu + 8\nu^2) \sinh^2(d\xi))\tilde{t}}{G\xi((3 - 4\nu) \cosh(2d\xi) + 2(d\xi)^2 + 5 - 12\nu + 8\nu^2)}, \quad (21)$$

which agrees with the classical result obtained for a two-dimensional thin elastic layer in the Fourier domain by Hannah<sup>[36]</sup>, and Eq. (20) reduces to

$$\tilde{u}_r(\xi, 0) = \frac{-((d\xi)^2 - (3 - 10\nu + 8\nu^2) \sinh^2(d\xi))\tilde{q} + (1 - \nu)(2d\xi + (3 - 4\nu) \sinh(2d\xi))\tilde{t}}{G\xi((3 - 4\nu) \cosh(2d\xi) + 2(d\xi)^2 + 5 - 12\nu + 8\nu^2)}. \quad (22)$$

In particular, the coefficient multiplying  $\tilde{q}$  in Eq. (21) exactly recovers the classical isotropic normal-traction kernel for a thin elastic layer with a fixed lower boundary. This is equivalent to Hannah's result for the vertical surface displacement (see Eq. (10))<sup>[36]</sup>. The same kernel of  $\tilde{q}$  was also written in the Fourier form as shown in Eq. (2.5) by Essink et al.<sup>[37]</sup>.

Although the exact expressions for the surface compliance functions are too cumbersome for direct analytical inversion, the inverse Hankel transform can be asymptotically evaluated in the limit of a long wavelength, i.e., when  $d\xi \ll 1$ . Specifically, Eq. (19) admits the expansion

$$\begin{aligned} \tilde{u}_z(\xi, 0)/d = & f_z^{(0)}\tilde{q} + f_z^{(1)}d\xi\tilde{t} + f_z^{(2)}d^2\xi^2\tilde{q} + f_z^{(3)}d^3\xi^3\tilde{t} + \cdots \\ & + f_z^{(n-1)}d^{(n-1)}\xi^{(n-1)}\tilde{t} + f_z^n d^n \xi^n \tilde{q}, \end{aligned} \quad (23)$$

where the expressions for the last two terms depend on whether  $n$  is odd or even; here, we present the form corresponding to an even  $n$ ; the coefficients  $f_z^{(n)}$  ( $n = 0, 1, 2, 3, \dots$ ) are given by

$$\begin{cases} f_z^{(0)} = \frac{1}{C_{33}}, \\ f_z^{(1)} = \frac{C_{44} - C_{13}}{2C_{33}C_{44}}, \\ f_z^{(2)} = \frac{C_{13}(C_{13} - C_{44})}{3C_{33}^2C_{44}}, \\ f_z^{(3)} = -\frac{5C_{13}(C_{13}^2 - C_{11}C_{33}) + C_{44}(4C_{13}C_{44} + 3C_{11}C_{33} - 7C_{13}^2)}{24C_{33}^2C_{44}^2}, \\ \vdots \end{cases} \quad (24)$$

For clarity, we retain the terms only up to a prescribed order. However, the expansion can be extended straightforwardly to an arbitrary order by continuing the asymptotic procedure, and the code for generating arbitrary-order terms is available online (see <https://github.com/14523GuozhengZhang/Asymptotic-models.git>). To recover the surface displacements in the physical space, we first rewrite the term involving  $\tilde{t}$  by converting the associated Hankel integral from order 1 to order 0. We then apply the Hankel inversion to Eq. (23), yielding

$$u_z(r, 0)/d = f_z^{(0)}q + f_z^{(1)}d \frac{d(rt)}{rdr} - f_z^{(2)}d^2 \frac{1}{r} \frac{d}{dr} \left( r \frac{dq}{dr} \right) - f_z^{(3)}d^3 \frac{1}{r} \frac{d}{dr} \left( r \frac{d}{dr} \left( \frac{d}{rdr}(rt) \right) \right) + \dots \quad (25)$$

Returning to the Cartesian coordinates, we denote the vertical displacement at the upper surface by  $w = u_z(r, 0)$  and the tangential traction vector acting on the surface by  $\mathbf{t} = (t_x, t_y)$ . It is then natural to extend Eq. (25) to non-axisymmetric loading by replacing the radial operators with their two-dimensional in-plane counterparts (gradient, divergence, and Laplacian). Accordingly, Eq. (25) can be generalized to

$$w = \frac{1}{C_{33}} dq + \frac{C_{44} - C_{13}}{2C_{33}C_{44}} d^2 \nabla \mathbf{t} - \frac{C_{13}(C_{13} - C_{44})}{3C_{33}^2C_{44}} d^3 \nabla^2 q. \quad (26)$$

To the best of our knowledge, this relationship has not been reported previously. In the isotropic case, it reduces to

$$w = \frac{(1 - 2\nu)d}{2G(1 - \nu)} q + \frac{(1 - 4\nu)d^2}{4G(1 - \nu)} \nabla \mathbf{t} - \frac{\nu(4\nu - 1)d^3}{6G(1 - \nu)^2} \nabla^2 q, \quad (27)$$

which is consistent with the result in Eq. (2.3) of Ref. [18], obtained via a systematic perturbation expansion of three-dimensional isotropic elasticity.

It should be emphasized that extending Eq. (25) from axisymmetric to non-axisymmetric loading, leading to Eq. (26), is necessary but not, by itself, sufficient. Nevertheless, the agreement observed in the isotropic limit is not accidental for two reasons: (i) both derivations are grounded in the same thin-layer asymptotic regime, and thus retain an identical hierarchy of terms in the displacement-traction relations; (ii) the coefficients obtained from the Hankel transform in polar axisymmetry coincide with those derived from the Fourier transform in the two-dimensional Cartesian coordinates, at least up to  $O(d^2\xi^2)$ , as evidenced by Eq. (21) and as noted by Hannah<sup>[36]</sup>. Beyond this formal correspondence, we further support the non-axisymmetric extension by validating the resulting reduced operators against the ones from the finite element method (FEM) reported later in this paper.

At the next order,  $O(d^3\xi^3)$ , caution is required. Specifically, for an axisymmetric in-plane traction with no azimuthal components, the two-dimensional curl vanishes, and thus

$$\nabla^2 \mathbf{t} = \nabla(\nabla \mathbf{t}) - \nabla^\perp(\text{curl } \mathbf{t}) = \nabla(\nabla \mathbf{t}).$$

Consequently, any cubic-order contribution that would distinguish between  $\nabla^2 \mathbf{t}$ ,  $\nabla(\nabla \mathbf{t})$ , or, equivalently, a term proportional to  $\nabla^\perp(\text{curl } \mathbf{t})$ , collapses to the same operator under axisymmetry, and the same collapse occurs for  $\mathbf{u}$ . This prevents a unique and principled extension of the  $O(d^3\xi^3)$  term from Eq. (25) to the fully non-axisymmetric case. For this reason, we restrict Eq. (26) to the terms up to  $O(d^2\xi^2)$ . We note that the reduced-order and long-wavelength formulation can be interpreted as a regular perturbation expansion in the small parameter  $d\xi$ , which represents the dimensionless in-plane wavenumber. It should also be noted that similar wavenumber-based asymptotic approaches have been used in nonlinear elastic problems, for instance, the surface-wrinkling analyses within the context of finite elasticity by Liu et al.<sup>[38–39]</sup>.

Similarly, we can perform an asymptotic expansion of Eq. (20) under the long-wavelength assumption  $d\xi \ll 1$ , obtaining

$$\tilde{u}_r(\xi, 0) = f_r^{(0)} d\tilde{t} + f_r^{(1)} d^2 \xi \tilde{q} + f_r^{(2)} d^3 \xi^2 \tilde{t} + f_r^{(3)} d^4 \xi^3 \tilde{q} + O((d\xi)^4), \quad (28)$$

where the coefficients  $f_r^{(n)}$  ( $n = 0, 1, 2, 3$ ) are

$$f_r^{(0)} = \frac{1}{C_{44}}, \quad f_r^{(1)} = f_z^{(1)}, \quad f_r^{(2)} = \frac{C_{13}^2 - C_{11}C_{33} - C_{44}(C_{13} - C_{44})}{3C_{33}C_{44}^2}, \quad f_r^{(3)} = f_z^{(3)}. \quad (29)$$

Transforming  $\tilde{q}$  to a first-order Hankel form and performing the inverse Hankel transform of order 1, we obtain the radial displacement at the surface as

$$u_r(r, 0) = f_r^{(0)} dt - f_r^{(1)} d^2 \frac{dq}{dr} - f_r^{(2)} d^3 \frac{d}{dr} \left( \frac{d}{rdr} (rt) \right) + f_r^{(3)} d^4 \frac{d}{dr} \left( \frac{1}{r} \frac{d}{dr} \left( r \frac{dq}{dr} \right) \right). \quad (30)$$

Also, Eq. (30) can be generalized to a non-axisymmetric loading condition in the Cartesian coordinates  $\mathbf{x} = (x, y)$  as

$$\mathbf{u} = \frac{1}{C_{44}} d\mathbf{t} - \frac{C_{44} - C_{13}}{2C_{33}C_{44}} d^2 \nabla q, \quad (31)$$

where  $\mathbf{u} = (u_x(\mathbf{x}, 0), u_y(\mathbf{x}, 0))$  denotes the tangential surface displacement, and  $\mathbf{t} = (t_x(\mathbf{x}), t_y(\mathbf{x}))$  is the tangential traction vector acting on the surface. This result becomes

$$\mathbf{u}(\mathbf{x}, 0) = \frac{d}{G} \mathbf{t} - \frac{(1 - 4\nu)d^2}{4G(1 - \nu)} \nabla q \quad (32)$$

for isotropic materials, which is also consistent with that in Eq. (2.3) of Ref. [18]. The second term on the right-hand side in Eq. (32) shows that the pressure-gradient-driven lateral flow within the bonded thin layer is itself controlled by the factor  $(1 - 4\nu)$ . When  $\nu > 1/4$ , this coupling reverses the sign, meaning that the same  $\nabla q$  produces an opposite contribution to the in-plane redistribution. Through the thin-layer kinematics and confinement, the divergence of this in-plane redistribution is precisely what feeds into the normal response, consistent with Eq. (27) where the same prefactor multiplies  $\nabla \mathbf{t}$ . Hence, the sign change at  $\nu = 1/4$  can be viewed as a compressibility-controlled reversal of how lateral redistribution couples back to lift-up versus sink-in.



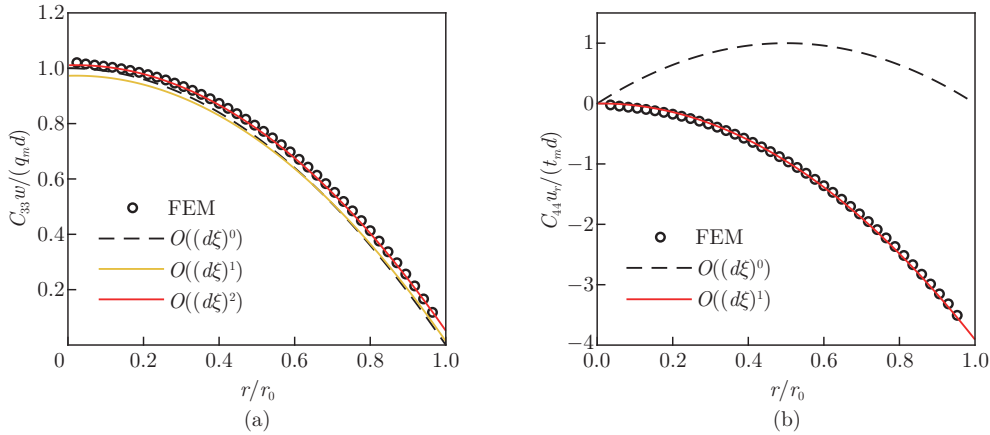
### 3.1.2 Numerical verifications

We then perform simulations using the FEM in ABAQUS to assess the consistency between the numerical results and the theoretical predictions. For the axisymmetric configuration, we model a deformable layer of radius  $r_0 = 20$  mm and thickness  $d = 1$  mm. The mesh uses CAX4R elements (4-node bilinear axisymmetric quadrilaterals with reduced integration) to reduce the computational cost. The material is considered as transversely isotropic and linearly elastic, with rubber-like stiffnesses  $(C_{11}, C_{12}, C_{13}, C_{33}, C_{44}, C_{66}) = (87.01, 76.97, 77.13, 86.86, 4.98, 5.02)$  MPa. Perfect bonding to a rigid substrate is represented by fully clamping the bottom surface (all translational and rotational degrees of freedom fixed), consistent with Eq. (18).

Spatially varying surface tractions are prescribed on the top surface. In the axisymmetric model, the radial shear traction and normal pressure are chosen as

$$t_r = 4t_m \left( \frac{r}{r_0} - \left( \frac{r}{r_0} \right)^2 \right), \quad q = q_m \left( 1 - \left( \frac{r}{r_0} \right)^2 \right) \quad (33)$$

for  $r < r_0$ , with the peak values  $t_m = 0.1$  Pa and  $q_m = 10$  Pa. The resulting surface displacements can be readily calculated by plugging Eq. (33) into Eqs. (26) and (31). Clearly, under these loading and boundary conditions, retaining higher-order terms systematically improves the agreement between the theoretical predictions and the FEM results for the top-surface displacements (see Fig. 2).

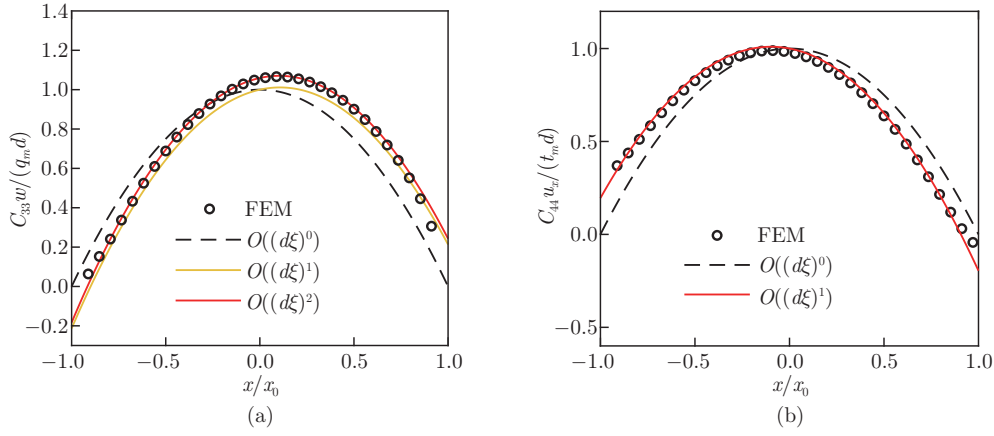


**Fig. 2** Comparison of the axisymmetric solutions with the FEM results: (a) top-surface deflection  $w(r)$  and (b) top-surface radial displacement  $u_r(r)$ , both plotted against the normalized radius  $r/r_0$ . Displacements are normalized as shown on the axes using the peak tractions  $q_m$  (normal),  $t_m$  (tangential), and the layer thickness  $d$ . Dashed and solid curves denote the solutions of different orders of the asymptotic expansion in  $d\xi$  (color online)

It is interesting to test the generalized equations (26) and (31) in a non-axisymmetric configuration. We then carry out a three-dimensional model of in-plane dimensions  $l_0 \times l_0$  with  $l_0 = 40$  mm and thickness  $d = 1$  mm, which are computed using C3D8R elements (8-node linear bricks with reduced integration). Again, the bottom surface is clamped in the same manner. Spatially varying tractions are applied in the Cartesian coordinates with peak values  $t_m = 1$  Pa and  $q_m = 4$  Pa,

$$t_x = t_m \left( 1 - \left( \frac{2x}{l_0} \right)^2 \right) \left( 1 - \left( \frac{2y}{l_0} \right)^2 \right), \quad t_y = 0, \quad q = q_m \left( 1 - \left( \frac{2x}{l_0} \right)^2 \right) \left( 1 - \left( \frac{2y}{l_0} \right)^2 \right) \quad (34)$$

for  $|x| < \frac{l_0}{2}$  and  $|y| < \frac{l_0}{2}$ . Analytical predictions readily obtained by plugging the surface traction into Eqs. (26) and (31) remain in good agreement with the three-dimensional FEM results for the top-surface displacements, particularly at higher orders (see Fig. 3).



**Fig. 3** Comparison of the asymptotic results with the FEM results in non-axisymmetric configurations: (a) top-surface deflection  $w(x)$  normalized by  $q_m d / C_{33}$  and (b) top-surface in-plane displacement  $u_x(x)$  normalized by  $t_m d / C_{44}$ , with  $q_m$  and  $t_m$  the peak normal and tangential tractions, respectively. Profiles along the centerline are plotted against the normalized coordinate  $x/x_0$ , where  $x_0 = l_0/2$  is half the model width. Dashed and solid curves correspond to successive truncations of the asymptotic expansion in  $d\xi$  (color online)

### 3.2 Prescribed surface displacements

We next consider a perfectly bonded elastic layer subject to prescribed surface displacements: the normal (vertical) displacement  $w(r)$ , whose zeroth-order Hankel transform is denoted by  $\tilde{w}(\xi)$ , and the radial (tangential) displacement  $u(r)$ , whose first-order Hankel transform is denoted by  $\tilde{u}(\xi)$ . The corresponding boundary conditions in the transform domain are expressed as

$$\begin{cases} \tilde{u}_z(\xi, 0) = \tilde{w}, & \tilde{u}_r(\xi, 0) = \tilde{u}, \\ \tilde{u}_z(\xi, -d) = 0, & \tilde{u}_r(\xi, -d) = 0. \end{cases} \quad (35)$$

Combining Eq. (35) with Eq. (16) can give rise to surface tractions in the Hankel domain in terms of the transformed surface displacements,

$$\tilde{q}(\xi) = \mathcal{K}_{zz}\tilde{w} + \mathcal{K}_{zr}\tilde{u}, \quad (36)$$

$$\tilde{t}(\xi) = \mathcal{K}_{rz}\tilde{w} + \mathcal{K}_{rr}\tilde{u}, \quad (37)$$

where  $\mathcal{K}_{ij}(\xi)$  ( $i, j = z, r$ ) are the stiffness kernels, and  $\mathcal{K}_{zr} = \mathcal{K}_{rz}$ .

By expanding Eq. (36) in the long-wavelength limit ( $d\xi \ll 1$ ), we obtain its asymptotic form as

$$\tilde{q}(\xi) = k_z^{(0)}\tilde{w}/d + k_z^{(1)}\xi\tilde{u} + k_z^{(2)}d\xi^2\tilde{w} + k_z^{(3)}d^2\xi^3\tilde{u} + O((d\xi)^4), \quad (38)$$

where the coefficients  $k_z^{(n)}$  ( $n = 0, 1, 2, 3$ ) are given by

$$\begin{cases} k_z^{(0)} = C_{33}, \\ k_z^{(1)} = \frac{1}{2}(C_{13} - C_{44}), \\ k_z^{(2)} = -\frac{(C_{13} + 3C_{44})(C_{13} - C_{44})}{12C_{44}}, \\ k_z^{(3)} = -\frac{(C_{13} + C_{44})(C_{11}C_{33} - C_{44}^2)}{24C_{33}C_{44}}. \end{cases} \quad (39)$$

Accordingly, by transforming back to the physical space, the generalized relation between the normal traction  $q$  and surface displacements  $(w, \mathbf{u})$  is

$$q = C_{33} \frac{w}{d} + \frac{C_{13} - C_{44}}{2} \nabla \mathbf{u} + \frac{(C_{13} + 3C_{44})(C_{13} - C_{44})}{12C_{44}} d \nabla^2 w. \quad (40)$$

The asymptotic expansion of Eq. (37) in the long-wavelength limit ( $d\xi \ll 1$ ) yields

$$\tilde{t}(\xi) = k_r^{(0)} \tilde{u}/d + k_r^{(1)} \xi \tilde{w} + k_r^{(2)} d \xi^2 \tilde{u} + k_r^{(3)} d^2 \xi^3 \tilde{w} + O((d\xi)^4), \quad (41)$$

where the coefficients are

$$k_r^{(0)} = C_{44}, \quad k_r^{(1)} = k_z^{(1)}, \quad k_r^{(2)} = \frac{4C_{11}C_{33} - 2C_{13}C_{44} - C_{13}^2 - C_{44}^2}{12C_{33}}, \quad k_r^{(3)} = k_z^{(3)}. \quad (42)$$

We then obtain the generalized tangential traction in terms of surface displacements in the real space as

$$\mathbf{t} = C_{44} \mathbf{u}/d - \frac{C_{13} - C_{44}}{2} \nabla w. \quad (43)$$

It is evident that  $k_z^{(0)} = 1/f_z^{(0)}$  and  $k_r^{(0)} = 1/f_r^{(0)}$ , implying that the models based on prescribed surface tractions (see Eqs. (26) and (31)) and those based on prescribed surface displacements (see Eqs. (40) and (43)) are identical at the leading order.

Notably, for an isotropic elastic layer, Eqs. (36) and (37) reduce to

$$\tilde{q}(\xi) = \frac{2(1-\nu)G\xi((3-4\nu)\sinh(2d\xi) + 2d\xi)\tilde{w} - 2G\xi((3-10\nu+8\nu^2)\sinh^2(d\xi) - (d\xi)^2)\tilde{u}}{(3-4\nu)^2\sinh^2(d\xi) - (d\xi)^2}, \quad (44)$$

$$\tilde{t}(\xi) = \frac{-2G\xi((3-10\nu+8\nu^2)\sinh^2(d\xi) - (d\xi)^2)\tilde{w} + 2(1-\nu)G\xi((3-4\nu)\sinh(2d\xi) - 2d\xi)\tilde{u}}{(3-4\nu)^2\sinh^2(d\xi) - (d\xi)^2}, \quad (45)$$

which yields a high-order Pasternak-type foundation model in the long-wavelength limit,

$$q = \frac{2(1-\nu)G}{(1-2\nu)d} w - \frac{(1-4\nu)G}{2(1-2\nu)} \nabla \mathbf{u} - \frac{(3-4\nu)(1-4\nu)Gd}{12(1-2\nu)^2} \nabla^2 w, \quad (46)$$

and a shear-lag model

$$\mathbf{t} = \frac{G}{d} \mathbf{u} + \frac{(1-4\nu)G}{2(1-2\nu)} \nabla w \quad (47)$$

for isotropic thin elastic layers. It is worth noting that, unlike the classical Winkler foundation and shear-lag models, the present high-order foundation relations (see Eqs. (40) and (46) for transversely isotropic and isotropic layers, respectively) and high-order shear-lag relations (see Eqs. (43) and (47), respectively) are intrinsically coupled. Specifically, the gradient of the tangential surface displacement contributes to the normal traction in Eqs. (40) and (46), while the gradient of the normal surface displacement enters the tangential traction in Eqs. (43) and (47). Such coupling arises naturally from the nonlocal nature of these models, and it becomes essential for accurately describing moderately thin layers.

We further verify the asymptotic results in this subsection by the FEM simulations under prescribed surface displacements on the top boundary. All geometric parameters, material properties, and boundary conditions for both the axisymmetric and three-dimensional models are identical to those used in the preceding subsection. The surface boundary conditions,

however, are changed to be prescribed displacements. For the axisymmetric model, the imposed radial displacement and deflection on the top surface are

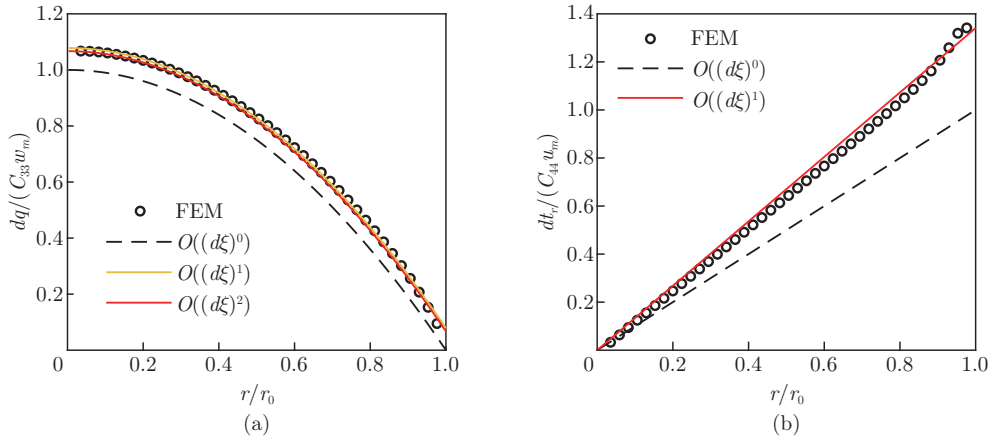
$$u_r = u_m \frac{r}{r_0}, \quad w = w_m \left(1 - \left(\frac{r}{r_0}\right)^2\right) \quad (48)$$

for  $r < r_0$ , where  $u_m = 10 \mu\text{m}$  and  $w_m = 5 \mu\text{m}$  are the peak values of the prescribed surface displacements.

The required surface tractions to achieve such displacements can be readily calculated by plugging Eq. (48) into Eqs. (40) and (43). Figure 4 shows that retaining higher-order terms improves the agreement between the theoretical predictions and the FEM results for the surface tractions. Similarly, we also verify the non-axisymmetric and three-dimensional model, for which spatially varying displacements are prescribed in the Cartesian coordinates as

$$u_x = u_m \left(1 - \left(\frac{2x}{l_0}\right)^2\right) \left(1 - \left(\frac{2y}{l_0}\right)^2\right), \quad u_y = 0, \quad w = w_m \left(1 - \left(\frac{2x}{l_0}\right)^2\right) \left(1 - \left(\frac{2y}{l_0}\right)^2\right) \quad (49)$$

for  $|x| < \frac{l_0}{2}$  and  $|y| < \frac{l_0}{2}$ . As shown in Fig. 5, the asymptotic results given by Eqs. (40) and (43) capture the surface tractions with good accuracy for asymmetric configurations.



**Fig. 4** Comparison of the predicted axisymmetric tractions on the top surface of the bonded elastic layer with the FEM results: (a) normal traction  $q(r)$  and (b) radial shear traction  $t_r(r)$ , plotted against the normalized radius  $r/r_0$  and scaled as indicated on the axes. Symbols denote the FEM results. Dashed and solid curves show successive truncations of the asymptotic expansion in  $d\xi$  (color online)

## 4 Frictionlessly bonded elastic layers

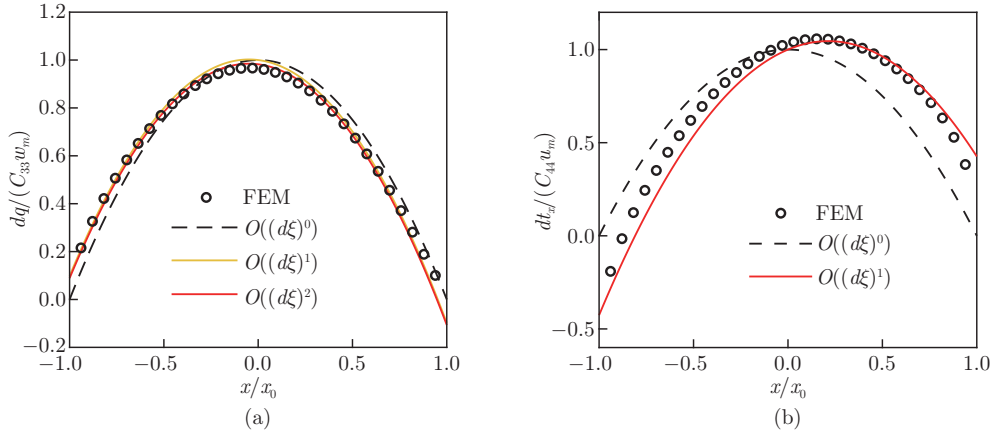
We next consider the case of a frictionlessly bonded elastic layer, motivated by the rolling or sliding contact problems discussed by Johnson<sup>[1]</sup>. In this situation, both the radial and tangential stress components vanish at the lower surface, while either the tractions or the displacements are prescribed at the upper surface.

### 4.1 Prescribed surface tractions

#### 4.1.1 Coupled and high-order solutions

Following the same notation as in the preceding subsection, the corresponding boundary conditions in the Hankel space for a frictionlessly bonded elastic layer subject to surface tractions are

$$\tilde{\sigma}_z(\xi, 0) = \tilde{q}, \quad \tilde{\sigma}_{rz}(\xi, 0) = \tilde{t}, \quad \tilde{u}_z(\xi, -d) = \tilde{\sigma}_{rz}(\xi, -d) = 0. \quad (50)$$



**Fig. 5** Non-axisymmetric tractions on the top surface of the bonded elastic layer: (a) normal traction  $q(x)$  and (b) shear traction  $t_x(x)$  in the  $x$ -direction, scaled as indicated on the axes. Centerline profiles are plotted against  $x/x_0$ , where  $x_0$  is half the model width. Symbols denote the FEM results, and dashed or solid curves correspond to successive truncations of the asymptotic expansion in  $d\xi$  (color online)

Combining Eq. (50) with Eq. (16) yields the following relationships between surface displacements and applied tractions:

$$\tilde{u}_z(\xi, 0) = \bar{\mathcal{F}}_{zz}(\xi)\tilde{q}(\xi) + \bar{\mathcal{F}}_{zr}(\xi)\tilde{t}(\xi), \quad \tilde{u}_r(\xi, 0) = \bar{\mathcal{F}}_{rz}(\xi)\tilde{q}(\xi) + \bar{\mathcal{F}}_{rr}(\xi)\tilde{t}(\xi), \quad (51)$$

where  $\bar{\mathcal{F}}_{ij}(\xi)$  ( $i, j = z, r$ ) are the compliance kernels for the frictionless interface.

In the same manner, the asymptotic expansion of the surface displacements for a frictionlessly bonded layer can be obtained as

$$\tilde{u}_z(\xi, 0) = \bar{f}_z^{(-1)}\tilde{t}/\xi + \bar{f}_z^{(0)}d\tilde{q} + \bar{f}_z^{(1)}d^2\xi\tilde{t} + O((d\xi)^2), \quad (52)$$

$$\tilde{u}_r(\xi, 0) = \bar{f}_r^{(-2)}\tilde{t}/(d\xi^2) + \bar{f}_r^{(-1)}\frac{1}{\xi}\tilde{q} + \bar{f}_r^{(0)}d\tilde{t} + \bar{f}_r^{(1)}d^2\xi\tilde{q} + O((d\xi)^2), \quad (53)$$

where the coefficients are

$$\begin{cases} \bar{f}_r^{(-2)} = \frac{C_{33}}{C_{11}C_{33} - C_{13}^2}, & \bar{f}_r^{(-1)} = \bar{f}_z^{(-1)} = -\frac{C_{13}}{C_{11}C_{33} - C_{13}^2}, \\ \bar{f}_z^{(0)} = \frac{C_{11}}{C_{11}C_{33} - C_{13}^2}, & \bar{f}_r^{(0)} = \frac{C_{11}C_{33} - 2C_{13}C_{44} - C_{13}^2}{3C_{44}(C_{11}C_{33} - C_{13}^2)}, \\ \bar{f}_z^{(1)} = \bar{f}_r^{(1)} = \frac{C_{11}}{3(C_{11}C_{33} - C_{13}^2)}. \end{cases} \quad (54)$$

Thus, we obtain the generalized models for the mechanical response of frictionlessly bonded layers,

$$\nabla^2 w = \frac{C_{13}}{C_{11}C_{33} - C_{13}^2} \nabla \mathbf{t} + \frac{C_{11}}{C_{11}C_{33} - C_{13}^2} d\nabla^2 q. \quad (55)$$

In this frictionless configuration, we set the applied surface shear traction to zero and focus on the resulting normal traction response; detailed discussion of the induced shear traction is therefore omitted. Hence, we prescribe only the normal load  $q$ , as shown in

$$\nabla \cdot \mathbf{u} = -\frac{C_{13}}{C_{11}C_{33} - C_{13}^2} q. \quad (56)$$

For isotropic layers, they reduce to

$$\nabla^2 w = \frac{\nu}{2G} \nabla \cdot \mathbf{t} + \frac{(1-\nu)d}{2G} \nabla^2 q, \quad (57)$$

$$\nabla \cdot \mathbf{u} = -\frac{\nu}{2G} q. \quad (58)$$

The forms presented in Eqs. (55) and (57) are reminiscent of the classical Kerr foundation model<sup>[11]</sup> and its subsequent modifications<sup>[12–13]</sup>, although the present formulation retains the additional coupling between normal and tangential responses. Notably, when the tangential traction vanishes ( $\mathbf{t} = \mathbf{0}$ ), the leading-order normal response in Eq. (57) reduces to

$$\frac{w}{d} = \frac{(1-\nu)}{2G} q,$$

which coincides with the result derived by Johnson<sup>[1]</sup> for a frictionless elastic layer based on ad hoc assumptions.

#### 4.1.2 Numerical verifications

In our simulation for the frictionlessly bonded case, the geometries, dimensions, and element types are identical to those used in the preceding section for the perfectly bonded case. The changes are the boundary conditions and material properties. Specifically, a frictionless support at the bottom surface is enforced by constraining the vertical displacement only (no penetration), i.e.,  $w = 0$  on the bottom boundary, consistent with Eq. (63). The layer is modeled as a transversely isotropic solid with reduced incompressibility (i.e., more compressible), with stiffnesses  $(C_{11}, C_{12}, C_{13}, C_{33}, C_{44}, C_{66}) = (14.06, 5.93, 6.16, 13.97, 4.29, 4.065)$  MPa.

To verify Eq. (55), spatially varying surface tractions are prescribed on the top surface. For the axisymmetric model, the radial shear traction and normal pressure are

$$t_r = 4t_m \left( \frac{r}{r_0} - \left( \frac{r}{r_0} \right)^2 \right), \quad q = q_m \left( 1 - \left( \frac{r}{r_0} \right)^2 \right) \quad (59)$$

for  $r < r_0$  with the peak values  $t_m = 1$  Pa and  $q_m = 10$  Pa. For the non-axisymmetric model, the applied tractions are

$$t_x = t_m \frac{2x}{l_0} \left( 1 - \left( \frac{2y}{l_0} \right)^2 \right), \quad t_y = 0, \quad q = q_m \left( 1 - \left( \frac{2x}{l_0} \right)^2 \right) \left( 1 - \left( \frac{2y}{l_0} \right)^2 \right) \quad (60)$$

for  $|x| < x_0 = \frac{l_0}{2}$  and  $|y| < \frac{l_0}{2}$ , with  $t_m = q_m = 1$  Pa. Figure 6 shows the asymptotic prediction for  $\nabla^2 w$  given by Eq. (55) in both axisymmetric and non-axisymmetric configurations. We find good agreement between our models and the FEM results, except in the region very near the edge, where the problem is inherently three-dimensional, and a boundary-layer analysis is required. Note that in these comparisons, the traction ratios are chosen as  $t_m:q_m = 1:10$  (axisymmetric) and 1:1 (three-dimensional) to better separate the contributions from different asymptotic orders.

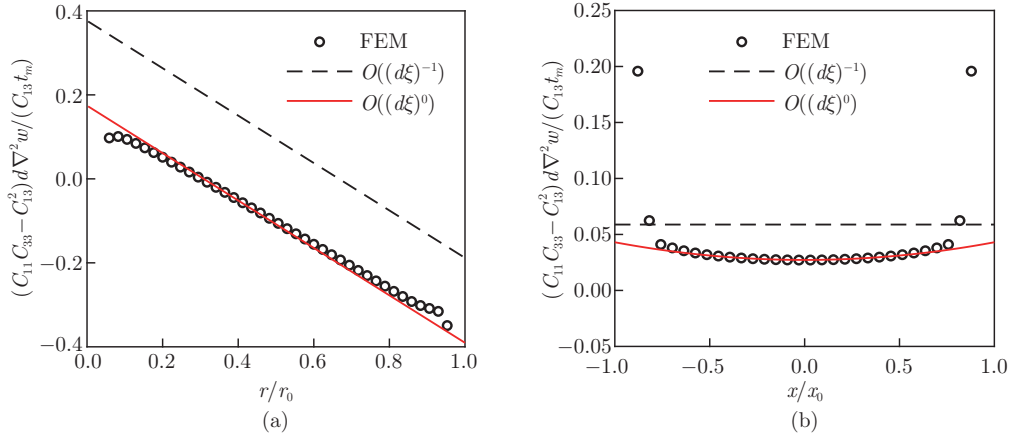
To verify the asymptotic prediction for  $\nabla \cdot \mathbf{u}$  given by Eq. (56), we prescribe a normal surface traction and set the shear traction to zero. Specifically, for the axisymmetric case, we apply

$$q = \left( 1 - 4 \left( \frac{r}{r_0} \right)^2 \right) q_m \quad (61)$$

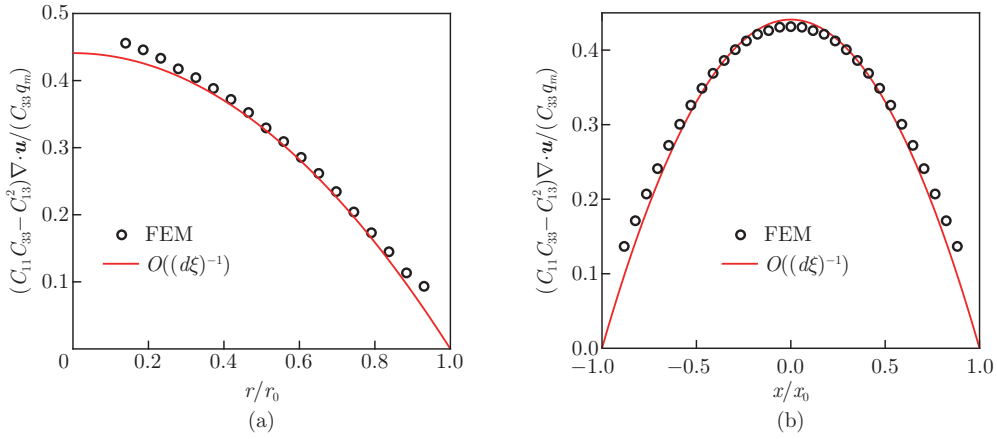
for  $r < \frac{r_0}{2}$ , and for the non-axisymmetric case,

$$q = q_m \left( 1 - 16 \left( \frac{x}{l_0} \right)^2 \right) \left( 1 - 16 \left( \frac{y}{l_0} \right)^2 \right) \quad (62)$$

for  $|x| < \frac{l_0}{4}$  and  $|y| < \frac{l_0}{4}$ . Such tractions are applied only over the inner portion of the domain to avoid edge-dominated responses. The comparison between the asymptotic theory and simulations is shown in Fig. 7. The asymptotic results from Eq. (56) accurately capture the surface tractions in both asymmetric and non-axisymmetric configurations.



**Fig. 6** Comparison of the theoretical and FEM results for  $\nabla^2 w$  under combined normal and tangential loading: (a) axisymmetric case plotted against  $r/r_0$  and (b) three-dimensional centreline results plotted against  $x/x_0$ . Symbols denote the FEM data, while dashed or solid curves correspond to successive truncations of the asymptotic expansion (color online)



**Fig. 7** Comparison of the asymptotic and FEM results for  $\nabla \cdot \mathbf{u}$ : (a) axisymmetric case and (b) non-axisymmetric case. Symbols denote the FEM data, while solid curves show the asymptotic prediction at the order indicated in the legend (color online)

#### 4.2 Prescribed surface displacements

We consider the case of prescribed surface displacements, for which the boundary conditions are

$$\tilde{u}_z(\xi, 0) = \tilde{w}, \quad \tilde{u}_r(\xi, 0) = \tilde{u}, \quad \tilde{u}_z(\xi, -d) = \tilde{\sigma}_{rz}(\xi, -d) = 0, \quad (63)$$

leading to the traction-displacement relations

$$\tilde{q}(\xi) = \bar{\mathcal{K}}_{zz} \tilde{w} + \bar{\mathcal{K}}_{zr} \tilde{u}, \quad \tilde{t}(\xi) = \bar{\mathcal{K}}_{rz} \tilde{w} + \bar{\mathcal{K}}_{rr} \tilde{u}, \quad (64)$$

where the stiffness kernels are given (see <https://github.com/14523GuozhengZhang/Asymptotic-models.git>). We then obtain the asymptotic form of Eq. (64) as

$$\begin{cases} \tilde{q} = \bar{k}_z^{(0)} \tilde{w}/d + \bar{k}_z^{(1)} \xi \tilde{u} + \bar{k}_z^{(2)} d \xi^2 \tilde{w} + \bar{k}_z^{(3)} d^2 \xi^3 \tilde{u} + O((d\xi)^4), \\ \tilde{t} = \bar{k}_r^{(1)} \xi \tilde{w} + \bar{k}_r^{(2)} d \xi^2 \tilde{u} + \bar{k}_r^{(3)} d^2 \xi^3 \tilde{w} + O((d\xi)^4), \end{cases} \quad (65)$$

where the coefficients are

$$\begin{cases} \bar{k}_z^{(0)} = C_{33}, & \bar{k}_z^{(1)} = \bar{k}_r^{(1)} = C_{13}, & \bar{k}_z^{(2)} = -\frac{C_{13}(C_{13} + 2C_{44})}{3C_{44}}, \\ \bar{k}_r^{(2)} = C_{11}, & \bar{k}_z^{(3)} = \bar{k}_r^{(3)} = -\frac{C_{11}(C_{13} + C_{44})}{3C_{44}}. \end{cases} \quad (66)$$

The corresponding generalized foundation and shear-lag relations for a frictionlessly bonded transversely isotropic elastic layer can then be written as

$$q = C_{33} \frac{w}{d} + C_{13} \nabla \cdot \mathbf{u}. \quad (67)$$

For an isotropic layer, Eq. (67) reduces to

$$q = \frac{2G(1-\nu)}{(1-2\nu)d} w + \frac{2G\nu}{1-2\nu} \nabla \cdot \mathbf{u}. \quad (68)$$

Note that as the bottom surface is frictionless, the present theory focuses on the normal traction response, and we therefore do not pursue a detailed analysis of the induced surface shear traction here.

Except for the loading protocol (i.e., prescribed surface displacements on the top boundary rather than applied tractions), all geometric parameters, mesh settings, and boundary conditions for both the axisymmetric and non-axisymmetric models are identical to those in the preceding subsection. Specifically, for the axisymmetric case, the imposed top-surface displacements are

$$u_r = 4u_m \left( \left( \frac{r}{r_0} \right)^2 - \frac{r}{r_0} \right), \quad w = \left( 1 - \left( \frac{r}{r_0} \right)^2 \right) w_m \quad (69)$$

for  $r < r_0$  with amplitudes  $u_m = 10 \mu\text{m}$  and  $w_m = 1 \mu\text{m}$ . For the non-axisymmetric case, the prescribed displacements are

$$u_x = -u_m \frac{2x}{l_0} \left( 1 - \left( \frac{2y}{l_0} \right)^2 \right), \quad u_y = 0, \quad w = w_m \left( 1 - 4 \left( \frac{x}{l_0} \right)^2 \right) \left( 1 - 4 \left( \frac{y}{l_0} \right)^2 \right) \quad (70)$$

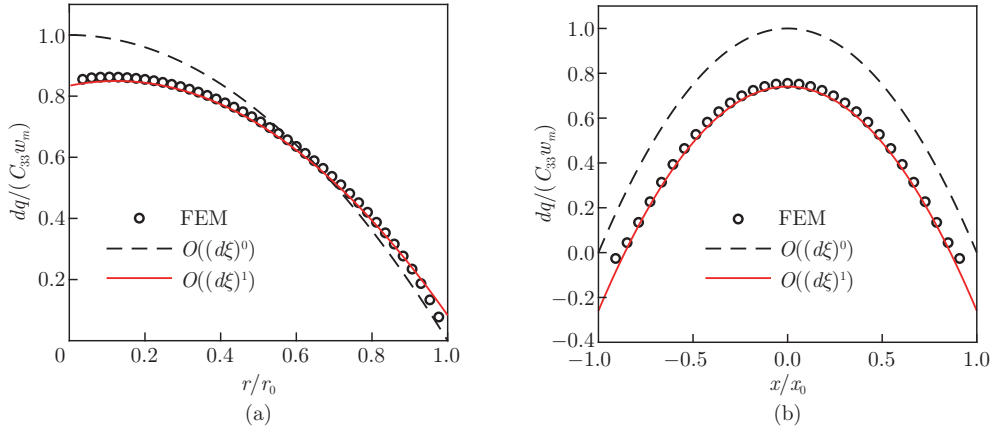
for  $|x| < x_0 = \frac{l_0}{2}$  and  $|y| < \frac{l_0}{2}$ . Figure 8 shows that, once terms up to  $O((d\xi)^1)$  are retained, the asymptotic prediction captures the FEM results for the induced normal traction  $q$  with good accuracy.

In many practical adhesive-layer settings, the bottom interface may be neither perfectly bonded nor perfectly frictionless, and is often modeled by Robin/spring-type constraints that relate tractions to interfacial slip or separation. This amounts to modifying the boundary-condition equations in our symbolic solver: a Robin-type interface can be implemented by setting the variables in `equ[]` in our code (see <https://github.com/14523GuozhengZhang/Asymptotic-models.git>) as  $Z = -k_n u_z$  and  $\tau = -k_t u_r$ . However, we focus on the perfectly bonded and frictionless limits, because they represent canonical extremes and enable transparent discussion of regime behavior, while retaining direct consistency with classical layer foundation treatments<sup>[3]</sup>.

## 5 High-order foundation and shear-lag models

Finally, we examine how the high-order and coupled thin-layer models derived in the preceding sections reduce to classical Winkler- and shear-lag-type formulations under specific boundary conditions. For example, consider the surface deflection  $w$  produced by a normal load  $q$





**Fig. 8** Comparison of the asymptotic and FEM results for the normal traction  $q$  induced by the coupled surface displacements: (a) axisymmetric case plotted against  $r/r_0$  and (b) non-axisymmetric case plotted against  $x/x_0$ . Symbols denote the FEM data, while dashed or solid curves correspond to successive truncations of the asymptotic expansion (color online)

when the top surface is free of shear traction, yielding an extended Winkler foundation model. Unlike the classical Winkler foundation, which assumes a purely local and uncoupled response, the present high-order formulation predicts a nonlocal relationship between  $q$  and  $w$ , as well as an additional shear deformation induced by the spatial gradients of the normal load. These nonlocal terms originate from higher-order contributions in the asymptotic expansion, specifically, the shear deformation within the layer, which becomes particularly significant for nearly incompressible materials.

### 5.1 Extended Winkler foundation models

We begin with the extended Winkler foundation models, in which the vertical surface displacement is expressed in terms of the applied normal load. The detailed coefficients in these relations, particularly those associated with higher-order terms, depend on the specified tangential boundary conditions. To elucidate their effects, we examine two limiting cases: perfectly bonded (no-slip) and frictionless (no-shear) interfaces at either the top or bottom surface of the layer.

As a representative case, when both the top and bottom surfaces are bonded to adjacent materials, referred to as no-slip (top) and no-slip (bottom), the boundary condition  $\mathbf{u}(\mathbf{x}, z = -d) = 0$  applies. Adopting the results derived in Subsection 3.2, we obtain the following relations between the surface deflection  $w$ , applied normal load  $q$ , and in-plane shear traction  $\mathbf{t}$ . For transversely isotropic layers,

$$w = \frac{d}{C_{33}}q - \frac{(C_{13} + 3C_{44})(C_{13} - C_{44})d^3}{12C_{33}^2 C_{44}}\nabla^2 q, \quad \mathbf{t} = \frac{(C_{44} - C_{13})d}{2C_{33}}\nabla q, \quad (71)$$

and for isotropic layers,

$$w = \frac{(1 - 2\nu)d}{2G(1 - \nu)}q + \frac{(3 - 4\nu)(1 - 4\nu)d^3}{48G(1 - \nu)^2}\nabla^2 q, \quad \mathbf{t} = \frac{(1 - 4\nu)d}{4(1 - \nu)}\nabla q. \quad (72)$$

These results clearly demonstrate that (i) the leading-order terms arise from volumetric (dilatational) deformation, whereas the second-order terms correspond to shape (shear) deformation, which becomes significant when the material is not perfectly compressible; (ii) the spatial gradients of the normal load can induce nonzero shear tractions under the no-slip bound-

ary condition, an effect neglected in classical and decoupled shear-lag models, where  $\mathbf{u} = \mathbf{0}$  is assumed to imply  $\mathbf{t} = \mathbf{0}$ .

Similarly, the mechanical response of the thin layer under the boundary conditions (no-shear (top) and no-slip (bottom)) can be readily obtained as

$$w = \frac{d}{C_{33}}q - \frac{C_{13}(C_{13} - C_{44})d^3}{3C_{33}^2C_{44}}\nabla^2q, \quad \mathbf{u} = \frac{(C_{13} - C_{44})d^2}{2C_{33}C_{44}}\nabla q \quad (73)$$

for transversely isotropic layers, and

$$w = \frac{(1 - 2\nu)d}{2G(1 - \nu)}q + \frac{\nu(1 - 4\nu)d^3}{6G(1 - \nu)^2}\nabla^2q, \quad \mathbf{u} = -\frac{(1 - 4\nu)d^2}{4G(1 - \nu)}\nabla q \quad (74)$$

for isotropic layers. Again, the gradient in  $q$  can also give rise to nonzero in-plane surface displacements.

In these relations from Eqs. (71)–(74), the terms of order  $O((d\xi)^0)$  are identical for different boundary conditions, whereas the higher-order terms differ owing to the distinct ability of the layers to sustain shear tractions. An exception arises for the boundary conditions, no-shear (top) and no-shear (bottom), under which the thin layer is always permitted to undergo volumetric deformation even when the material is perfectly incompressible. In this case, the surface response can be expressed as

$$w = \frac{C_{11}d}{C_{11}C_{33} - C_{13}^2}q, \quad \nabla \cdot \mathbf{u} = -\frac{C_{13}}{C_{11}C_{33} - C_{13}^2}q \quad (75)$$

for transversely isotropic layers, and

$$w = \frac{(1 - \nu)d}{2G}q, \quad \nabla \cdot \mathbf{u} = -\frac{\nu}{2G}q \quad (76)$$

for isotropic layers. Note that although the leading-order terms in these relations could be obtained from previous formulations based on either ad hoc assumptions or simplified loading conditions<sup>[25,29]</sup>, most of the shear-normal coupling terms (or higher-order corrections) derived here for transversely isotropic layers are, to the best of our knowledge, new (Eq. (73) is an exception, which has been partially reported in Eqs. (1.26) and (1.27) from Ref. [28]).

## 5.2 Extended Pasternak foundation models

We then proceed to the extended Pasternak foundation models, in which the restoring normal traction is expressed in terms of the prescribed vertical surface displacement. As before, two limiting interfacial conditions are considered: perfectly bonded (no-slip) and frictionless (no-shear) interfaces located at either the top or bottom surface of the layer. It is worth noting that the coefficients in the classical Pasternak foundation model are introduced phenomenologically<sup>[40]</sup>, whereas in the present formulation they emerge explicitly as functions of the elastic constants and thickness of the layer.

Specifically, under the no-slip (top) and no-slip (bottom) conditions, the surface tractions for a transversely isotropic layer are given by

$$q = \frac{C_{33}}{d}w + \frac{(C_{13} + 3CS_{44})(C_{13} - C_{44})d}{2C_{44}}\nabla^2w, \quad \mathbf{t} = \frac{C_{44} - C_{13}}{2}\nabla w, \quad (77)$$

which, for isotropic layers, reduce to

$$q = \frac{2(1 - \nu)G}{(1 - 2\nu)d}w - \frac{(3 - 4\nu)(1 - 4\nu)Gd}{12(1 - 2\nu)^2}\nabla^2w, \quad \mathbf{t} = \frac{(1 - 4\nu)G}{2(1 - 2\nu)}\nabla w. \quad (78)$$

Under the no-shear (top) and no-slip (bottom) conditions, the surface traction  $q$  and the displacement  $u$  are given by

$$q = \frac{C_{33}}{d}w + \frac{C_{13}(C_{13} - C_{44})d}{3C_{44}}\nabla^2 w, \quad \mathbf{u} = \frac{(C_{13} - C_{44})d}{2C_{44}}\nabla w \quad (79)$$

for transversely isotropic layers, and

$$q = \frac{2G(1 - \nu)}{(1 - 2\nu)d}w - \frac{2G\nu(1 - 4\nu)d}{3(1 - 2\nu)^2}\nabla^2 w, \quad \mathbf{u} = -\frac{(1 - 4\nu)d}{2(1 - 2\nu)}\nabla w \quad (80)$$

for isotropic layers. The first relationship in Eq. (79) coincides with Eq. (1.69) in Ref. [28], where, however, the corresponding coupling relation was not derived.

When both surfaces are frictionless, i.e., under the no-shear (top) and no-shear (bottom) conditions, the relationships are given by

$$q = \frac{C_{11}C_{33} - C_{13}^2}{C_{11}d}w, \quad \nabla \cdot \mathbf{u} = -\frac{C_{13}}{C_{11}d}w \quad (81)$$

for transversely isotropic layers, and

$$q = \frac{2G}{(1 - \nu)d}w, \quad \nabla \cdot \mathbf{u} = -\frac{\nu}{(1 - \nu)d}w \quad (82)$$

for isotropic layers.

### 5.3 Extended shear-lag models

We turn to the extended shear-lag models, in which the tangential surface traction is expressed in terms of the prescribed tangential surface displacement. The leading-order approximation remains accurate even for nearly incompressible materials, provided that the layer is sufficiently thin. In the following, we examine two representative normal surface conditions: zero surface deflection (no-deflection) and zero surface normal load (no-traction).

Under the no-deflection (top) and no-slip (bottom) conditions, the surface tractions for transversely isotropic layers are given by

$$\mathbf{t} = \frac{C_{44}}{d}\mathbf{u}, \quad q = \frac{C_{13} - C_{44}}{2}\nabla \cdot \mathbf{u}, \quad (83)$$

which, for isotropic layers, reduce to

$$\mathbf{t} = \frac{G}{d}\mathbf{u}, \quad q = -\frac{(1 - 4\nu)G}{2(1 - 2\nu)}\nabla \cdot \mathbf{u}. \quad (84)$$

Interestingly, the gradients of the surface shear displacement can induce nonzero normal restoring tractions, revealing the coupling between tangential and normal responses. This coupling is consistent with experimental observations in zero-degree peel tests, where out-of-plane deformation and even delamination arise under predominantly shear loading<sup>[41]</sup>.

Under the no-traction (top) and no-slip (bottom) conditions, the surface tractions and displacements for transversely isotropic layers are given by

$$\mathbf{t} = \frac{C_{44}}{d}\mathbf{u}, \quad w = \frac{(C_{44} - C_{13})d}{2C_{33}}\nabla \cdot \mathbf{u}, \quad (85)$$

which, for isotropic layers, reduce to

$$\mathbf{t} = \frac{G}{d} \mathbf{u}, \quad w = \frac{(1 - 4\nu)d}{4(1 - \nu)} \nabla \cdot \mathbf{u}. \quad (86)$$

## 6 Conclusions

Motivated by the incompleteness of asymptotic models for thin and transversely isotropic elastic layers, we establish a unified theoretical framework for the mechanics of thin and transversely isotropic elastic layers under general boundary conditions. Starting from the full three-dimensional elasticity equations, we systematically derive high-order surface traction-displacement relations through asymptotic expansion in the layer slenderness. Cases of more general conditions are complemented by a simple and spectral formulation using the Hankel transform. These approaches converge to yield consistent surface traction-displacement relations that explicitly capture the coupling between normal and tangential responses, generalizing the classical Winkler and shear-lag models to transversely isotropic materials of small thickness subject to a variety of surface/interface confinements.

The derived relations provide a rigorous route for computing the effective foundation stiffnesses directly from material constants. Such relations can bridge the gap between local, phenomenological spring-based descriptions and nonlocal, full elasticity solutions. It should be noted that the high-order terms discussed in this work reveal the physical origin of nonlocal compliance and shear-normal coupling, which should become increasingly significant for nearly incompressible materials or strongly anisotropic systems. Therefore, beyond reproducing and extending existing models, the present framework offers a versatile foundation for analyzing a wide range of problems involving bonded or coated layers, ranging from composite laminates and adhesive joints to soft thin films and biological tissues.

**Conflict of interest** The authors declare no conflict of interest.

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